

# Proxy Service - User Guide

## Proxy Service - User Guide

Complete guide for setting up, configuring, and using the Proxy Service.

### Table of Contents

1. [Installation](#)
2. [Configuration](#)
3. [Starting and Stopping](#)
4. [Client Setup](#)
5. [Android TV](#)
6. [VPN Setup](#)
7. [Cache Management](#)
8. [Monitoring](#)
9. [Advanced Configuration](#)
10. [FAQ](#)

### Quick Reference

- HTTP Proxy: `http://<HOST_IP>:53128`
  - HTTPS Proxy: `http://<HOST_IP>:53128`
  - SOCKS5 Proxy: `socks5://<HOST_IP>:51080`
  - Admin Panel: `http://<HOST_IP>:58080`
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## Installation

### Prerequisites

- **Container Runtime:** Docker or Podman
- **Compose Tool:** docker-compose, podman-compose, or built-in compose

- **VPN Config** (optional): .ovpn configuration file from your VPN provider

## Step 1: Clone Repository

```
git clone git@github.com:vasic-digital/Proxy.git
cd Proxy
```

## Step 2: Configure Environment

```
cp .env.example .env
nano .env # or your preferred editor
```

## Step 3: Initialize

```
./init
```

This creates:

- Configuration files
- Cache directories
- VPN auth file (if VPN enabled)

## Step 4: Start Service

```
./start
```

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# Configuration

## Basic Settings

Edit .env file:

```
# Network ports
HTTP_PROXY_PORT=53128      # HTTP/HTTPS proxy
SOCKS_PROXY_PORT=51080     # SOCKS5 proxy
PROXY_ADMIN_PORT=58080     # Admin web interface

# Cache settings
CACHE_DIR=/path/to/cache
CACHE_MAX_SIZE_GB=50
CACHE_MAX_AGE_DAYS=30

# Logging
LOG_LEVEL=info
LOG_DIR=./logs
```

## VPN Settings

```
USE_VPN=true
VPN_USERNAME=your_username
VPN_PASSWORD=your_password
VPN_OVPN_PATH=/path/to/config.ovpn
VPN_AUTO_RECONNECT=true
```

## Authentication

```
PROXY_AUTH_ENABLED=true
PROXY_USERNAME=your_proxy_user
PROXY_PASSWORD=your_proxy_pass
```

## Network Access Control

```
# Allowed networks (comma-separated CIDR)
ALLOWED_NETWORKS=192.168.0.0/16,10.0.0.0/8,172.16.0.0/12

# Bind to specific interface
BIND_ADDRESS=0.0.0.0 # All interfaces
# or
BIND_ADDRESS=192.168.1.100 # Specific IP
```

---

# Starting and Stopping

## Start Commands

```
# Normal start
./start

# Start without VPN
./start --no-vpn

# Verbose output
./start -v

# Pull latest images first
./start --pull

# Show status after start
./start --status
```

## Stop Commands

```
# Normal stop
./stop
```

```
# Stop and remove containers
./stop --remove

# Stop, remove containers and images
./stop --purge

# Stop and clear cache
./stop --clean-cache
```

## Restart Commands

```
# Normal restart
./restart

# Restart with cache clear
./restart --clean-cache

# Verbose restart
./restart -v
```

---

## Client Setup

### Finding Your Host IP

```
# Linux
ip addr show | grep "inet " | grep -v 127.0.0.1

# macOS
ifconfig | grep "inet " | grep -v 127.0.0.1
```

### Linux Clients

#### Temporary (Current Shell)

```
export HTTP_PROXY="http://192.168.1.100:53128"
export HTTPS_PROXY="http://192.168.1.100:53128"
export ALL_PROXY="socks5://192.168.1.100:51080"
```

#### Permanent (All Users)

Add to /etc/environment:

```
HTTP_PROXY="http://192.168.1.100:53128"
HTTPS_PROXY="http://192.168.1.100:53128"
NO_PROXY="localhost,127.0.0.1,.local"
```

## **APT (Debian/Ubuntu)**

Create /etc/apt/apt.conf.d/proxy.conf:

```
Acquire::http::Proxy "http://192.168.1.100:53128";  
Acquire::https::Proxy "http://192.168.1.100:53128";
```

## **YUM/DNF (RHEL/CentOS/Fedora)**

Add to /etc/dnf/dnf.conf or /etc/yum.conf:

```
proxy=http://192.168.1.100:53128
```

## **macOS Clients**

### **System-wide**

1. System Preferences → Network
2. Select network → Advanced → Proxies
3. Enable HTTP/HTTPS proxy
4. Web Proxy Server: 192.168.1.100:53128

### **Terminal (zsh)**

Add to ~/.zshrc:

```
export HTTP_PROXY="http://192.168.1.100:53128"  
export HTTPS_PROXY="http://192.168.1.100:53128"
```

## **Windows Clients**

### **System-wide**

1. Settings → Network & Internet → Proxy
2. Enable "Use a proxy server"
3. Address: 192.168.1.100, Port: 53128

### **PowerShell**

```
$env:HTTP_PROXY = "http://192.168.1.100:53128"  
$env:HTTPS_PROXY = "http://192.168.1.100:53128"
```

### **Command Prompt**

```
set HTTP_PROXY=http://192.168.1.100:53128  
set HTTPS_PROXY=http://192.168.1.100:53128
```

## Mobile Devices

### iOS

1. Settings → Wi-Fi → (network) → Configure Proxy
2. Select "Manual"
3. Server: 192.168.1.100, Port: 53128

### Android

1. Settings → Wi-Fi → (network) → Modify
2. Advanced options → Proxy → Manual
3. Hostname: 192.168.1.100, Port: 53128

## Browser Configuration

See [docs/BROWSERS.md](#) for detailed browser setup.

Quick configuration:

### Firefox

1. Settings → General → Network Settings → Settings
2. Select "Manual proxy configuration"
3. HTTP Proxy: 192.168.1.100, Port: 53128
4. SOCKS Host: 192.168.1.100, Port: 51080, Type: SOCKS v5
5. Check "Proxy DNS when using SOCKS v5"

### Chrome/Edge/Brave

Use system proxy settings or install SwitchyOmega extension:

1. Install SwitchyOmega from Chrome Web Store
  2. Create new profile → Proxy Profile
  3. HTTP: 192.168.1.100:53128
  4. SOCKS5: 192.168.1.100:51080
- 

## Android TV

Android TV is one of the most popular use cases for this proxy service. See the dedicated guide for complete setup instructions:

→ [\*\*Android TV Setup Guide\*\*](#)

Quick setup:

1. **Settings → Network & Internet → Wi-Fi**
2. Select your network → **Modify network**

### 3. **Advanced options** → Proxy: **Manual**

4. Proxy hostname: <HOST\_IP>

5. Proxy port: 53128

6. **Save**

If you see "Connected, no internet", the Squid proxy must be in standard forward proxy mode (not reverse/accel mode).  
Run `./init && ./restart` to regenerate the correct configuration.

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## VPN Setup

### Getting VPN Configuration

1. Subscribe to a VPN provider (NordVPN, ExpressVPN, Mullvad, etc.)
2. Download OpenVPN configuration files
3. Select a server location

### Configuration Steps

*# 1. Copy .ovpn file to project*

```
cp ~/Downloads/server.ovpn ./vpn-config.ovpn
```

*# 2. Edit .env*

```
nano .env
```

Add VPN settings:

```
USE_VPN=true
```

```
VPN_USERNAME=your_email@example.com
```

```
VPN_PASSWORD=your_vpn_password
```

```
VPN_OVPN_PATH=/absolute/path/to/vpn-config.ovpn
```

### Verify VPN Connection

*# Start with VPN*

```
./start
```

*# Check VPN status*

```
./status -v
```

*# Verify IP is masked*

```
curl --proxy http://localhost:53128 https://ifconfig.me
```

# VPN Troubleshooting

## Connection Fails

1. Check credentials
2. Verify .ovpn file path
3. Check container logs:

```
podman logs proxy-vpn
```

## DNS Leaks

Add to .env:

```
DNS_SERVERS=1.1.1.1,1.0.0.1
```

---

# Cache Management

## Viewing Cache Status

```
# Basic statistics
```

```
./cachectl stats
```

```
# Size breakdown
```

```
./cachectl size
```

```
# List cached content
```

```
./cachectl list
```

## Clearing Cache

```
# Clear with confirmation
```

```
./cachectl clear
```

```
# Force clear
```

```
./cachectl clear -f
```

## Manual Invalidation

```
# Remove stale files
```

```
./cachectl invalidate
```

```
# Trim to specific size
```

```
./cachectl trim 30 # 30 GB
```



## Automatic Invalidation

Configured in `.env`:

```
CACHE_AUTO_INVALIDATE=true  
CACHE_INVALIDATE_INTERVAL=24 # hours  
CACHE_MAX_AGE_DAYS=30
```

## Streaming Cache

```
# Enable streaming cache  
STREAMING_CACHE_ENABLED=true  
  
# Configure streaming domains  
STREAMING_DOMAINS=youtube.com,googlevideo.com,cdn  
  
# Streaming cache size  
STREAMING_CACHE_MAX_SIZE_GB=20
```

---

## Monitoring

### Service Status

```
# Basic status  
./status  
  
# Detailed status  
./status -v  
  
# JSON output (for scripting)  
./status --json  
  
# Continuous monitoring  
./status --watch
```

### Admin Panel

Access at: `http://HOST_IP:58080`

Features:

- Real-time service status
- Cache statistics
- Connection information
- Configuration overview

## Log Files

```
# Proxy logs
tail -f logs/proxy.log

# Squid access log
tail -f logs/squid/access.log

# Squid cache log
tail -f logs/squid/cache.log

# VPN log
podman logs -f proxy-vpn
```

## Health Checks

```
# HTTP proxy health
curl --proxy http://localhost:53128 http://
connectivitycheck.gstatic.com/generate_204

# SOCKS proxy health
curl --proxy socks5://localhost:51080 http://
connectivitycheck.gstatic.com/generate_204

# Admin health
curl http://localhost:58080/health
```

---

## Advanced Configuration

### Custom DNS

```
DNS_SERVERS=8.8.8.8,8.8.4.4,1.1.1.1
DNS_CACHING_ENABLED=true
DNS_CACHE_TTL=3600
```

### Performance Tuning

```
# Maximum concurrent connections
MAX_CONNECTIONS=4096

# Connection timeout
CONNECTION_TIMEOUT=60

# Cache memory
CACHE_MEMORY_SIZE_MB=512
```

## Ad Blocking

```
BLOCK_ADS=true  
BLOCK_MALICIOUS=true  
# Optional custom blocklist  
BLOCKLIST_FILE=./config/blocklist.txt
```

## Custom Squid Configuration

Edit config/squid/squid.conf for advanced settings:

```
# Add custom refresh patterns  
refresh_pattern -i youtube.* 10080 90% 43200 override-expire  
  
# Add custom ACLs  
acl mynetwork src 192.168.1.0/24  
http_access allow mynetwork
```

---

## FAQ

### Q: Can I use both HTTP and SOCKS proxies?

A: Yes, both run simultaneously on different ports. HTTP proxy on 53128, SOCKS on 51080.

### Q: How do I know if VPN is working?

A: Check your external IP:

```
curl --proxy http://localhost:53128 https://ifconfig.me
```

### Q: Why is caching not working for HTTPS?

A: HTTPS content is encrypted. Enable HTTPS inspection (MITM) for HTTPS caching, but this requires certificate setup.

### Q: How do I restrict access to specific IPs?

A: Set ALLOWED\_NETWORKS in .env:

```
ALLOWED_NETWORKS=192.168.1.0/24
```

### Q: Cache is growing too large

A: Reduce CACHE\_MAX\_SIZE\_GB or enable automatic invalidation:

```
CACHE_MAX_SIZE_GB=30  
CACHE_AUTO_INVALIDATE=true
```

## **Q: VPN disconnects frequently**

**A:** Enable auto-reconnect:

```
VPN_AUTO_RECONNECT=true  
VPN_HEALTH_INTERVAL=30
```

## **Q: Can I run this on a remote server?**

**A:** Yes. Update `BIND_ADDRESS=0.0.0.0` and configure firewall to allow access.

## **Q: How do I debug connection issues?**

**A:** Check logs and enable verbose mode:

```
LOG_LEVEL=debug  
./start -v  
tail -f logs/proxy.log
```

---

# **Additional Documentation**

- [Client Setup Guide](#) — All devices (consoles, smart TVs, NAS, etc.)
- [Android TV Guide](#) — Android TV / Google TV
- [Browser Guide](#) — Firefox, Chrome, Edge, Safari
- [Mobile Devices](#) — iOS and Android phones/tablets
- [Network Modes](#) — VPN vs no-VPN explained
- [Architecture](#) — System design
- [Caching](#) — Cache management
- [Troubleshooting](#) — Common issues

# **Support**

For issues and feature requests:

- GitHub Issues: <https://github.com/vasic-digital/Proxy/issues>
- Check docs/ directory for detailed documentation